

Fundamentals of Networking

- NETW191 Final Course Project
- **May 2024**
- **Barbara Espericueta**
- Prof. **Farooq Afzal**



Introduction



Fundamentals of Information Technology and Networking



Overall project: This course project covers the fundamentals of networking that play a vital role in the IoT world. The design and development process of building a network. This project includes SOHO router configurations, subnetting, connectivity testing, networking documentation, and SOHO wireless network security. In this project, we learn to emerge technology to build a network.



Objective: Learning of emerging technologies to build a network, including IoT.



Tools: aci learning, INFOSEC learning INC, Microsoft Visio, Google, live DeVry Study.

NETW191 Course Project

Module 2

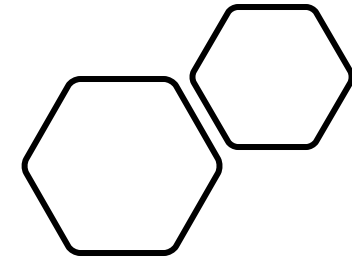
IPv4 Addressing

Introduction

This project is to implement an IPv4 addressing scheme to support a network. We will assign an IPv4 address to the SOHO Router virtual machine.

Summary

On Linux version, ubuntu computer 1. In the terminal we typed in the command `ip route | grep default`, this gives us the default IP address of the router we are interfacing with. Seen on screenshot Preparation. We then navigated to our Small Office/Home Office (SOHO) Router and changed the password for root by root. Now that we set a secure password, we navigated back to Computer 1, opened the computer's Web Browser, and typed in our IP address from our terminal to connect to our interface that is web-based for our router. We then applied changes to the Local Area Network (LAN), and now we have the desired IP address. Finally, we navigated back to our computer 1 terminal entered the `sudo ip link set eth0 down` to lose connection, and then `sudo ip set eth0 up` to regain connection. The connection is complete, now we can navigate back to the Web browser to see our successful connection. Seen in the screenshot IPv4 Address Assignment.





student@ubuntuvvm: ~/Desktop

```
student@ubuntuvvm:~/Desktop$ ip route | grep default
default via 192.168.1.1 dev eth0 proto dhcp metric 20100
student@ubuntuvvm:~/Desktop$
```

Preparation

- This screenshot should include the terminal window that shows **the default gateway IP address**.
- This screenshot should also show **the date/time information** on top of the Terminal window.

Activities Firefox Web Browser May 9 15:08

OpenWrt - Interfaces - LuX

192.168.105.1/cgi-bin/luci/admin/network/network

Status System Network Logout

Interfaces Global network options

Interfaces

LAN br-lan	Protocol: Static address Uptime: 0h 7m 55s MAC: 00:15:5D:00:BA:01 RX: 974.33 KB (11938 Pkts.) TX: 1.59 MB (11513 Pkts.) IPv4: 192.168.105.1/24 IPv6: fdf4:27c0:ac22::1/60	Restart Stop
TEST Alias of "lan"	Protocol: Alias Interface (Static address) Uptime: 0h 7m 55s IPv4: 192.168.100.1/24	Restart Stop Edit
WAN eth1	Protocol: DHCP client RX: 0 B (0 Pkts.) TX: 0 B (0 Pkts.) Error: Network device is not present	Restart Stop Edit

IPv4 Address Assignment

- This screenshot should include **the Interfaces page** that shows the new IPv4 address on the LAN interface.
- This screenshot should also show **the date/time information** next to the Firefox Web Browser tab.

Conclusion

We learned that a SOHO router is the core network hardware device used to build a small home or office LAN. Devices connected to the SOHO router via a cable or wirelessly are typically configured as DHCP clients by default. They get an IP address and other network configuration parameters from the SOHO router's DHCP server. Creating challenges in locating SOHO addresses. They cannot be used by hosts. The first valid host address will be assigned to the LAN interface of the SOHO Router. Locating this was a challenge, getting to know what to look for is crucial so getting familiar with locating addresses was a great skill to gain. In all we learned to implement an IPv4 addressing scheme to support a network and assign an IPv4 address to the SOHO Router virtual machine.

NETW191 Course Project

Module 3

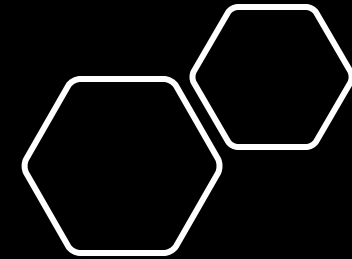
Connectivity Test

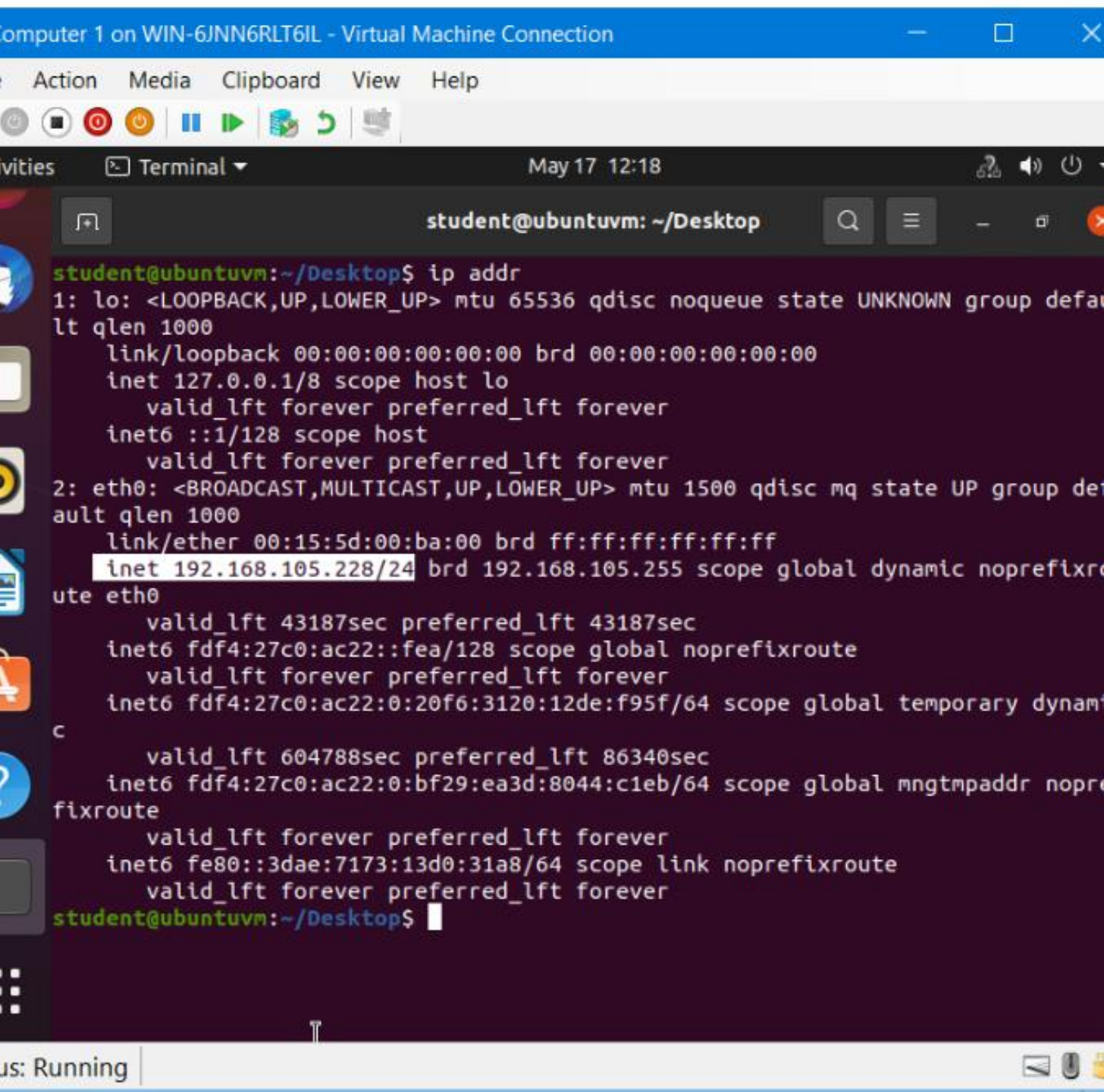
Introduction

This module aims to test the connectivity between the two computers and the SOHO Router.

Summary

A lot of functionality is built into a SOHO router. It contains a switch that supports multiple interfaces. It has routing functionality that connects your devices to the outside world (e.g., over a cable modem). It automatically performs network address translation between your LAN ports and the WAN port. Many SOHO routers include wireless access point functionality. The objective of this module is to test the connectivity between the two computers and the SOHO Router. First step, we typed *ip addr* in the terminal window to view the IP addresses of Computer 1 VM. Then we looked for the protocol family identifier *inet* when locating the VM's IPv4 address and found that it was 192.168.105.228/24, we see this in the first screenshot. The next step was to do the same on computer 2 and found that computer 2 IP address was 192.168.105.230/24. Now that Computer 1 VM, Computer 2 VM, and LAN port of the SOHO Router VM are on the same network. We used the ping utility program to verify the network connectivity among them. In the Terminal of Computer 1, we now verified connectivity to the SOHO Router VM (the default gateway) by using the following command *ping 192.168.105.1*. Same on Computer 2 Terminal with command *ping 192.168.105.228*. Verified connectivity was a success. Shown in screenshots 3 and 4.





```
student@ubuntuvm: ~/Desktop
student@ubuntuvm:~/Desktop$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:00:ba:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.105.228/24 brd 192.168.105.255 scope global dynamic noprefixroute eth0
        valid_lft 43187sec preferred_lft 43187sec
    inet6 fdf4:27c0:ac22::fea/128 scope global noprefixroute
        valid_lft forever preferred_lft forever
    inet6 fdf4:27c0:ac22:0:20f6:3120:12de:f95f/64 scope global temporary dynamic
        valid_lft 604788sec preferred_lft 86340sec
    inet6 fdf4:27c0:ac22:0:bf29:ea3d:8044:c1eb/64 scope global mngtmpaddr noprefixroute
        valid_lft forever preferred_lft forever
    inet6 fe80::3dae:7173:13d0:31a8/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
student@ubuntuvm:~/Desktop$
```

Dynamic IP Address Assignment

- Screenshot of the **Terminal window**.
- This screenshot should show the **IPv4 address of the Computer 1 VM**.
- This screenshot should also show the **date/time information** on top of the Terminal window.

Computer 2 on WIN-6JNN6RLT6IL - Virtual Machine Connection

File Action Media Clipboard View Help

Activities Terminal May 17 12:20

student@ubuntuvm: ~/Desktop

```
student@ubuntuvm:~/Desktop$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
lt qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group
ault qlen 1000
    link/ether 00:15:5d:00:ba:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.105.230/24 brd 192.168.105.255 scope global dynamic noprefi
ute eth0
        valid_lft 42727sec preferred_lft 42727sec
    inet6 fdf4:27c0:ac22::fea/128 scope global dadfailed tentative noprefixr
e
        valid_lft forever preferred_lft forever
    inet6 fdf4:27c0:ac22:0:9b58:1482:2354:16a5/64 scope global temporary dyn
c
        valid_lft 604786sec preferred_lft 85941sec
    inet6 fdf4:27c0:ac22:0:c0e2:9745:2c3a:4837/64 scope global mngtmpaddr no
fixroute
        valid_lft forever preferred_lft forever
    inet6 fe80::ac0f:8f1b:bc4f:9641/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
student@ubuntuvm:~/Desktop$
```

Status: Running

Dynamic IP Address Assignment

- Screenshot of the **Terminal window**.
- This screenshot should show the **IPv4 address of the Computer 2 VM**.
- This screenshot should also show the **date/time information** on top of the Terminal window.

Computer 1 on WIN-6JNN6RLT6IL - Virtual Machine Connection

File Action Media Clipboard View Help

Activities Terminal May 17 12:30

```
student@ubuntuvvm: ~/Desktop
64 bytes from 192.168.105.1: icmp_seq=11 ttl=64 time=0.337 ms
64 bytes from 192.168.105.1: icmp_seq=12 ttl=64 time=0.433 ms
64 bytes from 192.168.105.1: icmp_seq=13 ttl=64 time=0.347 ms
64 bytes from 192.168.105.1: icmp_seq=14 ttl=64 time=0.336 ms
64 bytes from 192.168.105.1: icmp_seq=15 ttl=64 time=0.335 ms
64 bytes from 192.168.105.1: icmp_seq=16 ttl=64 time=0.339 ms
^C
--- 192.168.105.1 ping statistics ---
16 packets transmitted, 16 received, 0% packet loss, time 15347ms
rtt min/avg/max/mdev = 0.278/0.325/0.433/0.040 ms
student@ubuntuvvm:~/Desktop$ ping 192.168.105.230
PING 192.168.105.230 (192.168.105.230) 56(84) bytes of data.
64 bytes from 192.168.105.230: icmp_seq=1 ttl=64 time=0.616 ms
64 bytes from 192.168.105.230: icmp_seq=2 ttl=64 time=0.411 ms
64 bytes from 192.168.105.230: icmp_seq=3 ttl=64 time=0.338 ms
64 bytes from 192.168.105.230: icmp_seq=4 ttl=64 time=0.382 ms
64 bytes from 192.168.105.230: icmp_seq=5 ttl=64 time=0.496 ms
64 bytes from 192.168.105.230: icmp_seq=6 ttl=64 time=0.410 ms
64 bytes from 192.168.105.230: icmp_seq=7 ttl=64 time=0.397 ms
64 bytes from 192.168.105.230: icmp_seq=8 ttl=64 time=0.394 ms
64 bytes from 192.168.105.230: icmp_seq=9 ttl=64 time=0.410 ms
64 bytes from 192.168.105.230: icmp_seq=10 ttl=64 time=0.380 ms
64 bytes from 192.168.105.230: icmp_seq=11 ttl=64 time=0.358 ms
64 bytes from 192.168.105.230: icmp_seq=12 ttl=64 time=0.404 ms
^C
--- 192.168.105.230 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11265ms
rtt min/avg/max/mdev = 0.338/0.416/0.616/0.070 ms
student@ubuntuvvm:~/Desktop$
```

Status: Running

Connectivity Test

- Screenshot of the **Terminal window**.
- This screenshot should show 1) the connectivity test **FROM the *Computer 1* VM TO the *SOHO Router* VM**, and 2) the connectivity test **FROM the *Computer 1* VM TO the *Computer 2* VM**.
- This screenshot should also show **the date/time information** on top of the Terminal window.


```
valid_lft forever preferred_lft forever
student@ubuntuvm:~/Desktop$ ping 192.168.105.1
PING 192.168.105.1 (192.168.105.1) 56(84) bytes of data:
64 bytes from 192.168.105.1: icmp_seq=1 ttl=64 time=0.313 ms
64 bytes from 192.168.105.1: icmp_seq=2 ttl=64 time=0.337 ms
64 bytes from 192.168.105.1: icmp_seq=3 ttl=64 time=0.362 ms
64 bytes from 192.168.105.1: icmp_seq=4 ttl=64 time=0.326 ms
64 bytes from 192.168.105.1: icmp_seq=5 ttl=64 time=0.332 ms
64 bytes from 192.168.105.1: icmp_seq=6 ttl=64 time=0.305 ms
64 bytes from 192.168.105.1: icmp_seq=7 ttl=64 time=0.288 ms
64 bytes from 192.168.105.1: icmp_seq=8 ttl=64 time=0.329 ms
^C
--- 192.168.105.1 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7148ms
rtt min/avg/max/mdev = 0.288/0.324/0.362/0.020 ms
student@ubuntuvm:~/Desktop$ ping 192.168.105.228
PING 192.168.105.228 (192.168.105.228) 56(84) bytes of data:
64 bytes from 192.168.105.228: icmp_seq=1 ttl=64 time=0.336 ms
64 bytes from 192.168.105.228: icmp_seq=2 ttl=64 time=0.511 ms
64 bytes from 192.168.105.228: icmp_seq=3 ttl=64 time=0.428 ms
64 bytes from 192.168.105.228: icmp_seq=4 ttl=64 time=0.494 ms
64 bytes from 192.168.105.228: icmp_seq=5 ttl=64 time=0.536 ms
64 bytes from 192.168.105.228: icmp_seq=6 ttl=64 time=0.440 ms
64 bytes from 192.168.105.228: icmp_seq=7 ttl=64 time=0.502 ms
^C
--- 192.168.105.228 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6138ms
rtt min/avg/max/mdev = 0.336/0.463/0.536/0.063 ms
student@ubuntuvm:~/Desktop$
```

Connectivity Test

- Screenshot of the **Terminal window**.
- This screenshot should show 1) the connectivity test **FROM the Computer 2 VM TO the SOHO Router VM**, and 2) the connectivity test **FROM the Computer 2 VM TO the Computer 1 VM**.
- This screenshot should also show **the date/time information** on top of the Terminal window.

Conclusion

The objective of this module was to test the connectivity between the two computers and the SOHO Router. Challenges faced in this project, locating the IP address. Confusion with inet as a location for the IP address. A little confusion gave skills in implementing better performance and security.

NETW191 Course Project

Module 4

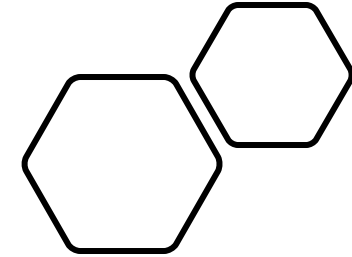
IP Subnetting and Loopback Interfaces

Introduction

In this module, we started with a class C network and divided its block of IP addresses into smaller blocks (i.e., subnets). Next, we configured two Loopback interfaces on the SOHO Router, using IP addresses of newly created subnets, to simulate two LAN segments.

Summary

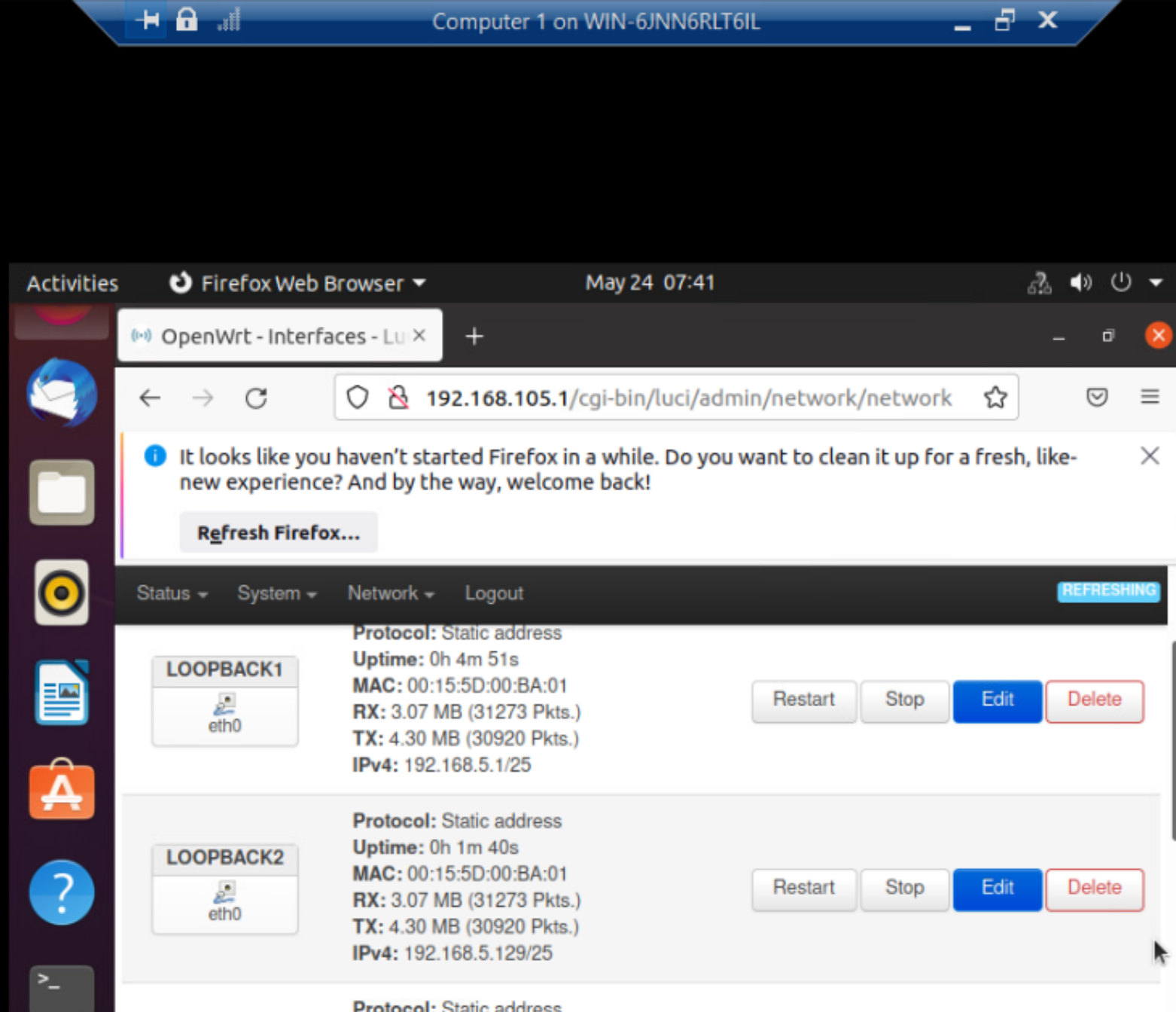
We started with a class C network and divided its block of IP addresses into smaller blocks (i.e., subnets). Next, we configured two Loopback interfaces on the SOHO Router, by using IP addresses of newly created subnets, to simulate two LAN segments. We were given an IP address block 192.168.5.0/24(network mask 255.255.255.0) and were asked to divide this block of addresses into two equal-sized subnets. This requires one bit to be borrowed from the host portion. When the borrowed (subnet) bit is 0, it represents the first subnet. When the borrowed (subnet) bit is 1, it represents the second subnet. With the borrowed bit being added to the network portion, both subnets have a prefix /25 or network mask 255.255.255.128. We already know /25 is 255.255.255.128. Due to changing our net mask into binary numbers. 11111111.11111111.11111111.10000000. Now that we have our network mask and our network addresses, we will be able to plug in the broadcast addresses and the first and last usable hosts. Now that we have all our information, we established a connection and tested out ping in the terminal shown in screenshots 1 and 2.



IP Subnetting

- This table should include **two /25 subnets**, listing
 - Subnet notation
 - Network address
 - First usable host address
 - Last usable host address
 - Broadcast address

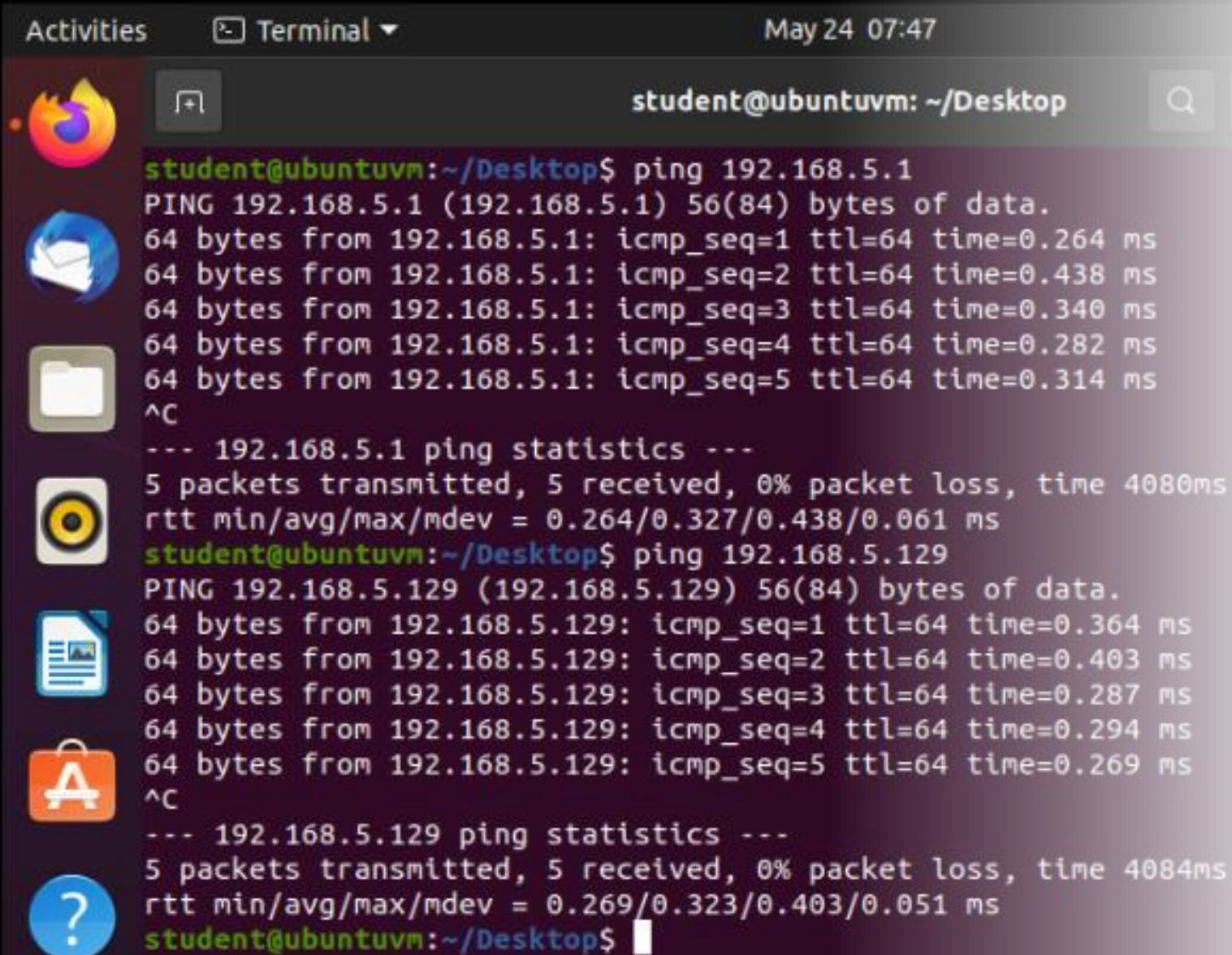
Subnet ID	Network Mask (/prefix)	Network Mask (Dotted decimal)	Network Address	First Usable Host Address	Last Usable Host Address	Broadcast Address
0	/25	255.255.255.128	192.168.5.0	192.168.5.1	192.168.5.126	192.168.5.127
1	/25	255.255.255.128	192.168.5.128	192.168.5.129	192.168.5.254	192.168.5.255



Loopback Interfaces

- This screenshot should show **both Loopback1 and Loopback2 interfaces** and their correct **IPv4 addresses**.
- This screenshot should also show **the date/time information** on top of the desktop window.

Connectivity Tests



The screenshot shows a terminal window titled "student@ubuntuvm: ~/Desktop" with a search icon and a menu icon. The terminal output shows two ping tests. The first test is to 192.168.5.1, showing 5 successful pings with times ranging from 0.264 ms to 0.438 ms. The second test is to 192.168.5.129, also showing 5 successful pings with times ranging from 0.269 ms to 0.403 ms. Both tests show 0% packet loss and a total time of approximately 4080ms and 4084ms respectively. The terminal window is part of a desktop environment with a sidebar on the left containing icons for Firefox, a mail client, a file manager, a music player, a document viewer, an application store, and a help icon. The top of the window shows "Activities", "Terminal", and the date "May 24 07:47".

```
student@ubuntuvm: ~/Desktop
student@ubuntuvm:~/Desktop$ ping 192.168.5.1
PING 192.168.5.1 (192.168.5.1) 56(84) bytes of data.
64 bytes from 192.168.5.1: icmp_seq=1 ttl=64 time=0.264 ms
64 bytes from 192.168.5.1: icmp_seq=2 ttl=64 time=0.438 ms
64 bytes from 192.168.5.1: icmp_seq=3 ttl=64 time=0.340 ms
64 bytes from 192.168.5.1: icmp_seq=4 ttl=64 time=0.282 ms
64 bytes from 192.168.5.1: icmp_seq=5 ttl=64 time=0.314 ms
^C
--- 192.168.5.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4080ms
rtt min/avg/max/mdev = 0.264/0.327/0.438/0.061 ms
student@ubuntuvm:~/Desktop$ ping 192.168.5.129
PING 192.168.5.129 (192.168.5.129) 56(84) bytes of data.
64 bytes from 192.168.5.129: icmp_seq=1 ttl=64 time=0.364 ms
64 bytes from 192.168.5.129: icmp_seq=2 ttl=64 time=0.403 ms
64 bytes from 192.168.5.129: icmp_seq=3 ttl=64 time=0.287 ms
64 bytes from 192.168.5.129: icmp_seq=4 ttl=64 time=0.294 ms
64 bytes from 192.168.5.129: icmp_seq=5 ttl=64 time=0.269 ms
^C
--- 192.168.5.129 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4084ms
rtt min/avg/max/mdev = 0.269/0.323/0.403/0.051 ms
student@ubuntuvm:~/Desktop$
```

- This screenshot should show **two successful ping tests**:
 - **from** the Computer 1 VM **to** the Loopback 1 interface
 - **from** the Computer 1 VM **to** the Loopback 2 interface
- This screenshot should also show **the date/time information** on top of the desktop window.

Conclusion

We started with a class C network and divided its block of IP addresses into smaller blocks (i.e., subnets). Next, we configured two Loopback interfaces on the SOHO Router, by using IP addresses of newly created subnets, to simulate two LAN segments. Challenges I faced during this project were dividing the block of addresses into two equal-sized subnets. Not knowing we borrowed a bit to do so, this strengthened my skill in subnetting, organization with subnetting tables, and with connectivity testing.

NETW191 Course Project

Module 5

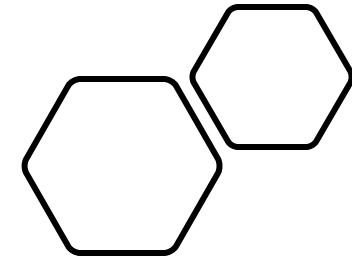
Visio Network Diagram

Introduction

In this module, we used Microsoft Visio to create a network diagram that depicts the interconnection of the Computer 1 VM, the Computer 2 VM, and the SOHO Router VM.

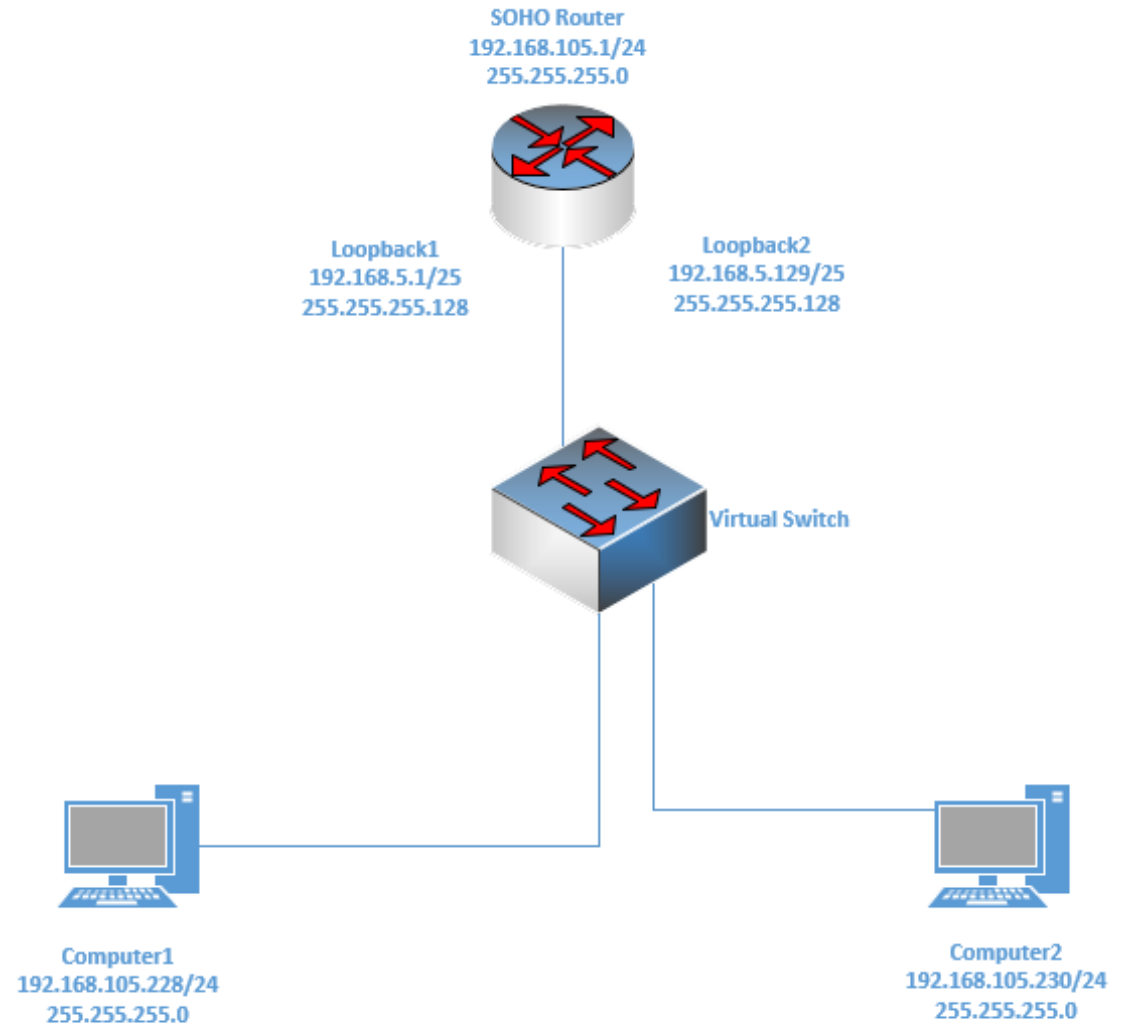
Summary

In Course Project Module 3, we performed a connectivity test between two computers and an SOHO Router and in Course Project Module 4, we configured two Loopback interfaces on the SOHO Router, by using IP addresses of newly created subnets, to simulate two LAN segments. In this module, we created a network diagram that depicts the interconnection of the Computer 1 VM, the Computer 2 VM, and the SOHO Router VM using Microsoft Visio. Microsoft Visio is a diagramming and vector graphics application that lets you transform hard-to-understand text and tables into organized visual diagrams. In our Visio Network Diagram, we created a diagram using everything we have been working on throughout our modules, creating a well-organized and understandable visual diagram of our network.



Microsoft Visio Network Diagram

- This diagram should illustrate **the interconnection** of the Computer 1 VM, the Computer 2 VM, and the SOHO Router VM along with proper **IP addresses** and **labeling**.
- This diagram should also show **your initials** in the bottom right corner.



Conclusion

In our Visio Network Diagram, we created a diagram using everything we have been working on throughout our modules, creating a well-organized and understandable visual of our network. During this project, a challenge I faced, locating different types of wiring. Learning to provide a comprehensive view of data so others would know what I was creating was a challenge. Using Visio created new skills in Microsoft. Having the ability to communicate ideas through visual diagrams sets skills in quickly identifying problems to quickly developing solutions. In a work environment improving organizational efficiency is crucial and is a great skill to have in cybersecurity.

NETW191 Course Project

Module 6

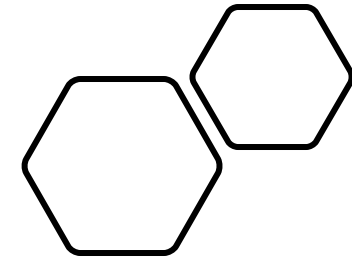
SOHO Wireless Network Security

Introduction

In this module, we explore some router configurations that could be used to add protection to a SOHO wireless network.

Summary

In this project module, we explored router configurations that could be used to add protection to a SOHO wireless network. We visited <https://www.tp-link.com/us/support/emulators>. The emulator is a virtual web GUI where you can experience the TP-Link product management. A great place to learn how to secure networks. We first selected our hardware and were shown the default username and password. A security concern, we are shown how vulnerable defaults are, as they can be searched for by anyone through websites like <https://www.routerpasswords.com/>. We then navigated to DHCP Settings, to the client's list where we were given the choice to select how we would configure our IP addresses. Either through configuring IP addresses manually (static IP) or using a DHCP server (dynamic IP). Now in device management, we navigated to the Wireless MAC Filter, showing MAC addresses that were being filtered by either having the administrator allow access to deny access to the network. Right back to the left navigation pane, we navigated to Wireless 2.4 GHz, Wireless Security. When given other choices for network security we are strongly encouraged to select WPA2-PSK AES (Advanced Encryption Standard) encryption as AES unites with multiple rounds of encryption making the network not worth hacking.



SOHO Wireless Network Security 1 of 3

1. What are the factory default username and password of a TP-Link router? Why is it important to change the default username and password of a SOHO router?

Answer: The default username is **admin** and the password is **admin**. Changing the default username and password is important because your router's default information is public knowledge. Anyone can find the default username and password to a router on a website like TP-Link or ROUTER Passwords.com.

2. To protect a SOHO wireless network with a small number of devices, which address management method provides more control, configuring the device IP addresses manually (static IP) or using a DHCP server (dynamic IP)? Why?

Answer: The method that provides more control in protecting your SOHO wireless network is to configure the device IP addresses manually (Static IP). With Static IP, Network admins manually configure IP addresses making them more secure, DHCP can be difficult to track due to the IP addresses being assigned by default.

SOHO Wireless Network Security 2 of 3

3. What does MAC filtering do? If needed, when would you use deny filtering rules, and when would you use allow filtering rules? What happens to devices that want to connect, if the “Allow the stations specified by any enabled entries in the list to access” function is enabled but there are no entries in the list?

Answer: MAC filtering takes the MAC addresses connected to a network, your laptop, computers, smartphones, routers, etc. For anything that connects to your network, MAC filtering will pick up on the MAC address, and depending on what your table addresses are assigned. The MAC filtering will either allow access or deny access to the network, addresses can be blocked as well. If needed when letting a device on the network is allowed by the administrator, but the administrator does not want to allow access anymore, the administrator can apply MAC filtering the deny filtering rule to deny access. If there are no entries to the list, then no devices will be able to connect. Since there are no entries including your computer and the function is enabled, you would have to manually connect via ethernet cable and include your MAC address to the filtering list or turn the feature off temporarily so you can add addresses.

4. What wireless security settings are displayed on the Wireless Security page? Which one is recommended by the vendor? Why?

Answer: On the wireless security page, it is clearly stated “For network security, it is strongly recommended to enable wireless security and select WPA2-PSK AES (Advanced Encryption Standard) encryption. AES unites with multiple rounds of encryption making the network not worth hacking for it will take way too long.

SOHO Wireless Network Security 3 of 3

5. Among the configurations you explored in this module, which one is a true security function? Why?

Answer: The true security function I would argue would be the WPA2-PSK AES, due to it incorporating multiple rounds of encryption for outsiders trying to get in. Encryption protects data, making it unreadable to unauthorized people and entities, scrambling and protecting our information, I would argue a true security function.

6. What would you do to protect your wireless network at home? Why?

Answer: What I would do to protect my wireless network at home would be to keep updating my router firmware and create a guest network for my children or visitors for protection from vulnerable websites and malware.

Conclusion

In this project module, we explored router configurations that are used to add protection to a SOHO wireless network. All these techniques can be used to make a SOHO network more secure. Changing the default username and password so that malicious users can not access your network is a huge step everyone should take. A challenge that I faced in this module was the reality of not knowing how important securing our network is and the steps to do so. Starting in my home with my router was my first step in securing my network. Routing configurations helped me gain skills in filtering and setting conditions in a network. These qualities are essential due to networks having large open ports for security threats, this skill would help companies tremendously.



Challenges/Lessons Learned

A challenge I faced during this project was dividing the block of addresses into two equal-sized subnets. How I overcame this challenge was I looking at various IP subnets and kept practicing until I felt confident. A lesson I learned during this project is practice makes perfect.

Career Skills Gained



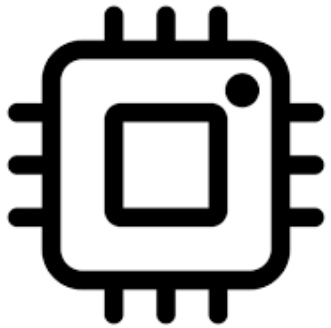
Networking



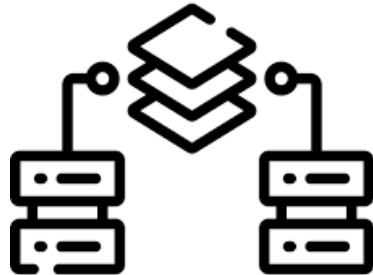
Troubleshooting



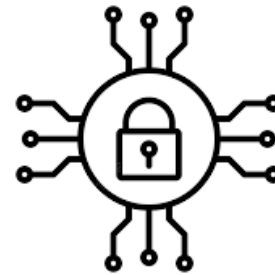
Cloud Computing



Hardware



Virtualization



Network Security

Conclusion

Progress in Networking has been rapid in recent years, and do not expect it to slow down anytime soon. This foundation of emerging technologies, including IoT has enormous future potential. As the workplace becomes digitalized, success is fueled by the ability to use technology to make data-driven decisions. This makes having skills in Information Technology and Networking essential to everyone. This course project covers the fundamentals of networking that play a vital role in the IoT world. Building a network's design and development process includes SOHO router configurations, subnetting, connectivity testing, networking documentation, and SOHO wireless network security.

References

<https://www.tp-link.com/us/support/emulator/>

<https://www.routerpasswords.com/>

Rasmussen, Anders Ingeman, A. (2010). *Microsoft Visio 2010 – Open source Visio alternatives.*

<https://www.osalt.com/visio>